

Subject:	IDATT1002 Software Engineering Project Assignment	
Class:	BIDATx 2023	Work category: Project
Teachers:	Surya Kathayat Grethe Sandstrak Muhammad Ali Norozi	
Date of issue:	30.01.2023	
First iteration:	03.03.2023	
Second iteration	24.03.2023	
Deadline: (Third iteration)	28.04.2023	
Presentation:	Week 18, 2023	



Image by pressfoto on Freepik



Image by Racool_studio on Freepik

Project Assignment

Your task is to create a tool for private individuals or small businesses to get an overview of income and expenses and help to improve the economy. The application should aid budgeting as well as accounting.

Each group is responsible to identify and make agreements with persons who can take the roles and responsibilities as user(s) and product owner (customer).

The requirements explained above are not final. Additional functions and features should be added according to feedback from the product owner and user tests.

Technical requirements

1. The application must be made as a standalone Java application (desktop application)
2. Keep persistent data in local files (as presented in Programming 2) or databases

Process

The project should be carried out in three iterations. The results of the two first iterations are presented in the first two team meetings, while the result of the third is presented in the project presentation. Iteration overview:

1. In the 1st iteration you should focus on the vision and requirements. Make an early prototype using BalsamiQ and perform a usability test using the prototype with users outside of the team. You should present this prototype at the first team meeting with the lecturer along with the vision document. You should also present the first version of a use case diagram and the domain model, which is essential in explaining the problem.
2. In the 2nd iteration you should focus on making an MVP (Minimum Viable Product) using Java. MVP is a minimal viable product, meaning, it does not need to have more functionality than the bare minimum that is needed for the product to be usable. As such, what functionality that there is, should be relatively free of bugs, which is what in general differentiates an MVP from a prototype. This MVP should also be presented to the users and tested. You should present this MVP at the second team meeting with the lecturer along with the first version of the requirements documentation WIKI.
3. In the 3rd iteration you should focus on finishing the application and all the documentation and attachments.

Basic project elements

Your team is employed by a computer consulting company that has been commissioned to develop the new information system. Notice that the team must create its own full-fledged system. As a student, you will mainly play the role of a systems consultant designing and implementing the system. Lecturer plays the role as a mentor in addition to expert advisors.

- As a systems consultant, you have an hourly salary rate of NOK 1599.-.
- You must consider any costs in terms of personnel and software.
- Each team member has approx. 120 hours for her/his disposal +/- 10 %.
- In the milestone plan, you will find deadlines for compulsory deliveries.
- To ensure the quality of the application the team must continuously undertake unit testing using JUnit.

- To ensure high usability and a good user experience the application must relate to Don Norman's principles of interaction design
- In order to evaluate the user experience the team must carry out usability tests after the first and second iteration (wireframe and MVP)
- Universal Design – the application must be designed according to WCAG 2.1 principle 1 - Perceivable

Resources

You will find the project resources, including document templates, in the project-folder on Blackboard.

Usages of collaboration tools

The team must use collaboration tools as a part of the project. E.g. tools like Gitlab (Trondheim, Ålesund - https://gitlab.stud.idi.ntnu.no/users/sign_in, Gjøvik - https://git.gvk.idi.ntnu.no/users/sign_in), GitLab WIKI, GitLab Issueboards, Gitlab Pages, BalsamiQ, Google Drive etc. Experiences in using these tools must be summarized in the main report. Balsamiq license: Available in Blackboard.

NB!

- Name your gitlab-project after following format: **idatx1002_2023_groupno**
- You must add the lecturer to your Gitlab repository as **Reporter**.
- Include the full name of all team-members on the Wiki-landing page.

Submission of reports

Each team must deliver a separate main report as well as attachments (

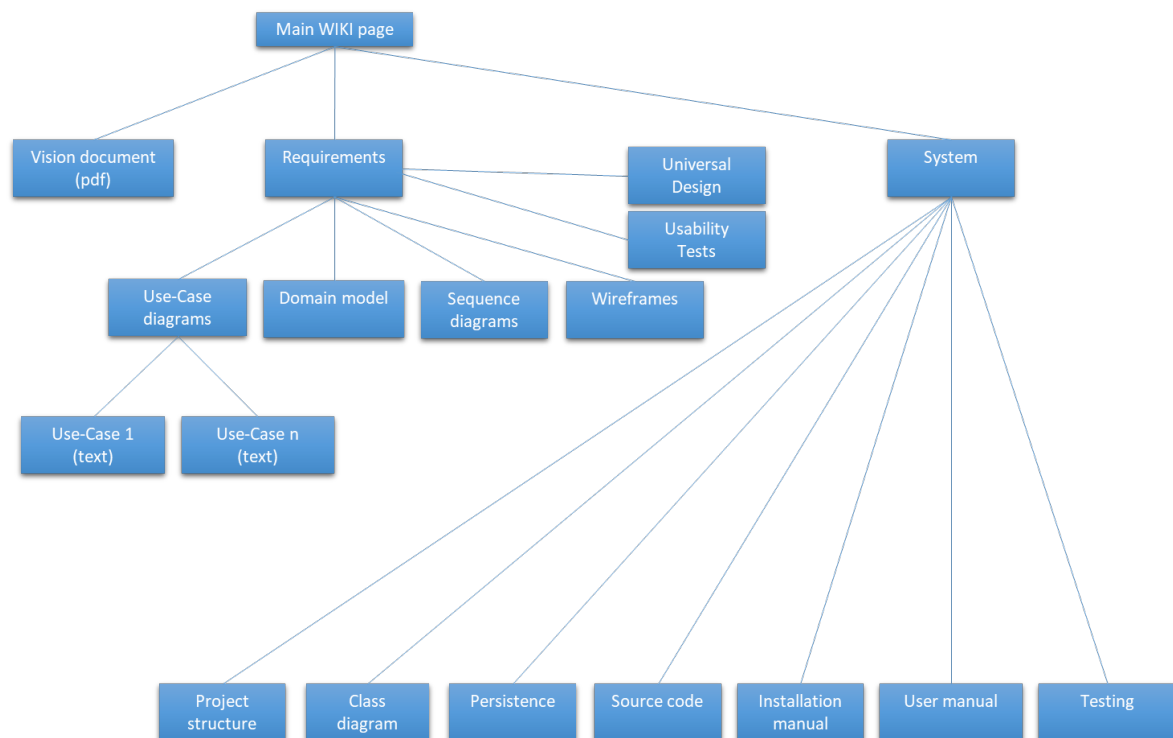
see below). The main report should include an evaluation of the teamwork. Here, the team must summarize experiences dealing with the project-work and cooperation within the team and with their product owner if applicable. Each team member must also document and evaluate his/her own performance.

Keywords are:

- This is how the teamwork progressed
- How was effort and result in accordance to what was planned
- This is what we are most pleased with
- We would have done this differently
- Source code or technical solutions the team is particularly satisfied with. Explain why
- Describe the choices and experiences (positive / negative) about the use of collaborative tools

Attachments to the Main report:

1. Project Manual
 - a. Collaboration agreement
 - b. Project plan with the schedule in the form of a Gantt chart
 - c. Timesheets with status reports for each team member
 - d. Meeting invitations and minutes
 - e. Gitlab issueboard, screenshots from each iteration
2. Vision document
3. Link to GitLab WIKI pages with the following content (NB! See lecture on documentation for a more specific description of each content element):



4. Link to JavaDoc on GitLab Pages

Final team deliverable

Inspira submissions:

1. Final report (pdf-file named `idata1002_2023_Report_groupno.pdf`)
2. Project manual, vision document, gitlab issuebord and WIKI pages (all appendix's included into one single pdf named `idata1002_2023_appendix_groupno.pdf`)
3. Source code + javadoc (zip file named `idata1002_2023_code_groupno.zip`)
4. URL – GitLab repository

Final individual deliverable

A self-reflection report prepared individually and uploaded on blackboard in which you reflect on your contribution to the group, what you think went well and not well from your side and how the team functioned.

- Your name and Team number.
- Describe the team-process as you perceive it (what went well or not)
- Describe your contribution and role in this project
- Analyze your teams process and your own contribution and role (why do you think the team process became as you describe it?)
- Looking forward - if you find yourself in a similar situation - based on the experiences from this project how would you act? Experience gained in this project: will it impact your further studies?
- Conclusion - What have you learned (learning outcomes), are there elements from this project you would want to explore even further. Your overall opinion of the team, process and project.